ALY 6110 · Big Data & Data Management

Seattle 911 CAD Call Data

Dataset Overview & Exploratory Analysis

GROUP	MEMBERS	INSTRUCTOR	TERM
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Data Source: City of Seattle Open Data Portal · data.seattle.gov/Public-Safety/Call-Data/33kz-ixgy

WHY THIS DATASET · Rationale for Selection

The Seattle CAD Call Data was selected for three interconnected reasons: public safety relevance, big data scale, and analytical richness.

01 Public Safety Relevance

Emergency response efficiency directly impacts community safety. Understanding patterns in 911 call volume, dispatch priority, and response time enables the Seattle Police Department to allocate officers more effectively, reduce response delays, and improve outcomes for residents.

02 Big Data Scale & Complexity

At 5.5 GB and over 10 million records spanning 2009–2026, the dataset genuinely requires big data techniques — chunked loading, distributed processing, and outlier filtering — making it ideal for a Big Data & Data Management course project.

03 Multi-Dimensional Analytical Potential

The dataset captures temporal, geographic, categorical, and operational dimensions simultaneously. This richness allows EDA across time-of-day, day-of-week, precinct geography, call classification, and resolution outcomes — five distinct analytical angles from one source.

04 Open Access & Reproducibility

Published on Seattle's Open Data Portal and updated regularly, the dataset is freely available, well-documented, and reproducible — meeting academic integrity standards and enabling future longitudinal comparisons as new years are added.

05 Predictive Modeling Opportunity

Dispatch priority, call type, timestamps, and resolution codes together provide enough signal to build a practical response-time prediction model.

DATASET OVERVIEW

About the Dataset

The Seattle CAD (Computer-Aided Dispatch) Call Data is published by the City of Seattle Open Data Portal. It records every 911 call received by the Seattle Police Department.

Coverage spans 2009–2026, capturing call classification, dispatch priority, precinct assignment, timestamps, geographic coordinates, and resolution outcomes.

At approximately 5.5 GB, the dataset required big data techniques including chunked loading, deduplication, and outlier filtering before analysis.

Key Data Columns

CAD Event Number
Unique identifier per event

Initial / Final Call Type
Classification of the incident

Priority
Dispatch urgency level

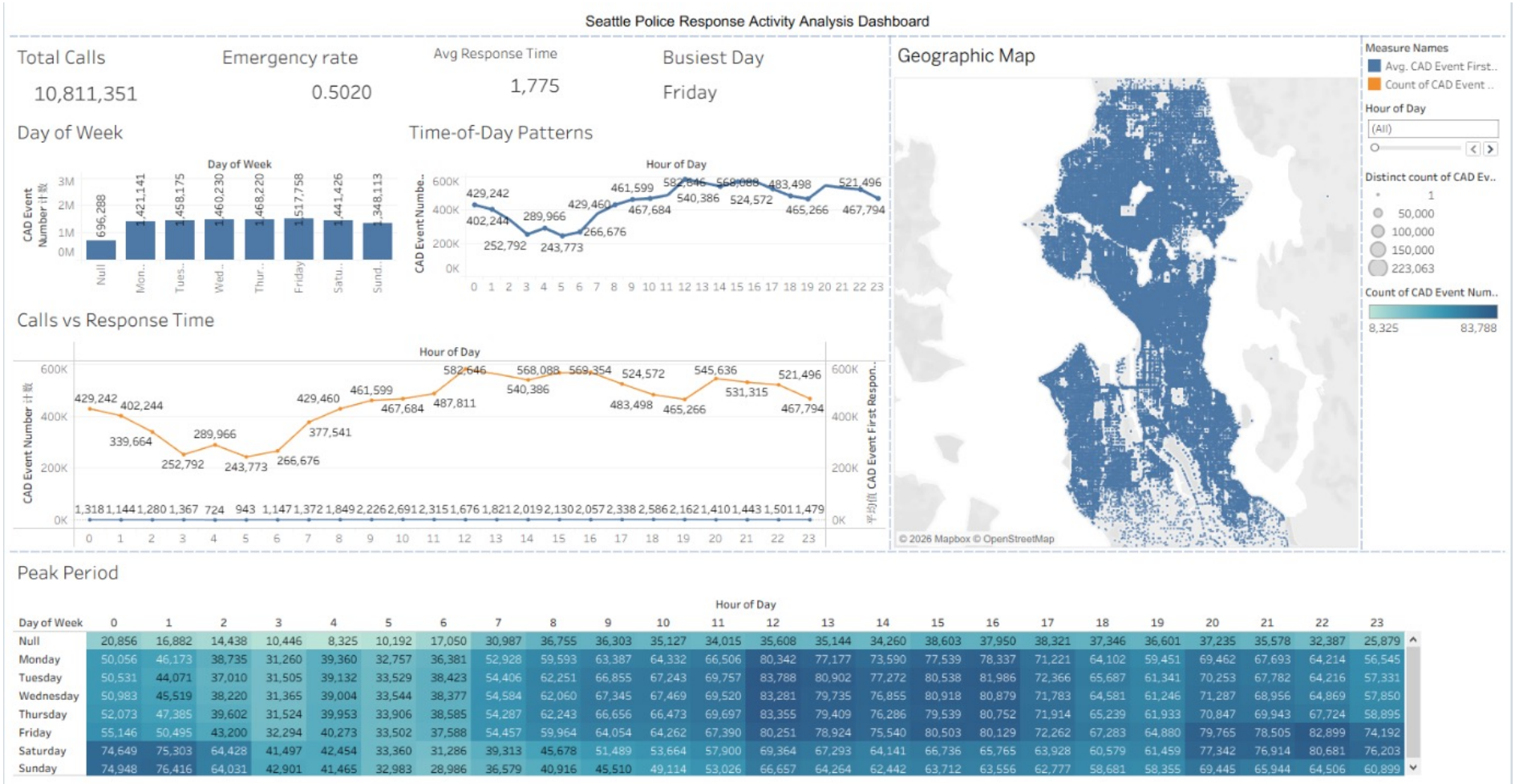
Precinct / Sector / Beat
Geographic policing unit

Timestamps
Incident datetime & response time

Clearance Description
Resolution outcome of the call

Longitude / Latitude
Exact geographic coordinates

DASHBOARD 1 · Seattle Police Response Activity Analysis



Peak Period

Day of Week	Hour of Day																							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Null	20,856	16,882	14,438	10,446	8,325	10,192	17,050	30,987	36,755	36,303	35,127	34,015	35,608	35,144	34,260	38,603	37,950	38,321	37,346	36,601	37,235	35,578	32,387	25,879
Monday	50,056	46,173	38,735	31,260	39,360	32,757	36,381	52,928	59,593	63,387	64,332	66,506	80,342	77,177	73,590	77,539	78,337	71,221	64,102	59,451	69,462	67,693	64,214	56,545
Tuesday	50,531	44,071	37,010	31,505	39,132	33,529	38,423	54,406	62,251	66,855	67,243	69,757	83,788	80,902	77,272	80,538	81,986	72,366	65,687	61,341	70,253	67,782	64,216	57,331
Wednesday	50,983	45,519	38,220	31,365	39,004	33,544	38,377	54,584	62,060	67,345	67,469	69,520	83,281	79,735	76,855	80,918	80,879	71,783	64,581	61,246	71,287	68,956	64,869	57,850
Thursday	52,073	47,385	39,602	31,524	39,953	33,906	38,585	54,287	62,243	66,656	66,473	69,697	83,355	79,409	76,286	79,539	80,752	71,914	65,239	61,933	70,847	69,943	67,724	58,895
Friday	55,146	50,495	43,200	32,294	40,273	33,502	37,588	54,457	59,964	64,054	64,262	67,390	80,251	78,924	75,540	80,503	80,129	72,262	67,283	64,880	79,765	78,505	82,899	74,192
Saturday	74,649	75,303	64,428	41,497	42,454	33,360	31,286	39,313	45,678	51,489	53,664	57,900	69,364	67,293	64,141	66,736	65,765	63,928	60,579	61,459	77,342	76,914	80,681	76,203
Sunday	74,948	76,416	64,031	42,901	41,465	32,983	28,986	36,579	40,916	45,510	49,114	53,026	66,657	64,264	62,442	63,712	63,556	62,777	58,681	58,355	69,445	65,944	64,506	60,899

DASHBOARD 1 · Key Insights

Calls by Day of Week

Friday receives the most calls (517,758). Volume increases steadily Mon–Fri, then drops on weekends — directly mirroring urban work-week activity cycles.

Hourly Pattern

Lowest volume at 3–6 AM. Volume builds through the morning, peaking midday (~582K at hour 11–15). Declines after 10 PM, consistent with daily human activity rhythms.

Calls vs. Response Time

Dual-axis chart confirms positive correlation: as hourly call volume rises, average CAD first response time also increases — indicating capacity strain at peak periods.

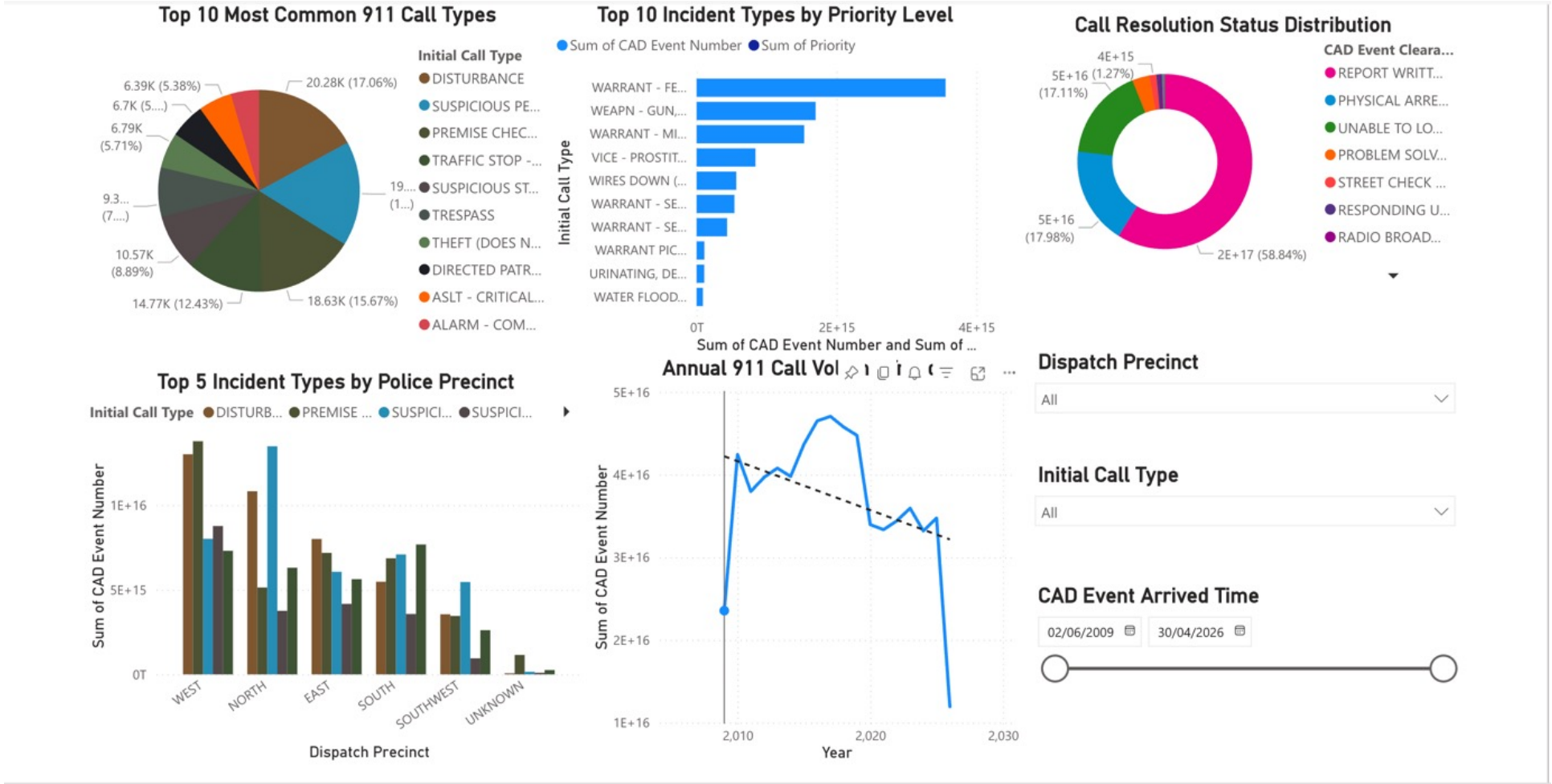
Weekday × Hour Heatmap

Peak cell: Tuesday 12 PM. Trough: Sunday 6 AM. The heatmap directly supports staff scheduling decisions to prevent under- or over-deployment at specific time slots.

Geographic Event Map

CAD events cluster densely in central Seattle. Outer suburban areas show significantly lower density, consistent with population concentration patterns citywide.

DASHBOARD 2 · Incident Types, Priority & Resolution Analysis



DASHBOARD 2 · Key Insights

Top 10 Call Types (Pie)

DISTURBANCE leads at 17.06%, followed by SUSPICIOUS PERSON (15.67%) and PREMISE CHECK (12.43%). Together, the top three categories account for nearly 45% of all calls.

Priority by Call Type

WARRANT and WEAPON calls receive highest dispatch priority. Parking violations rank lowest. The system effectively differentiates life-threatening emergencies from routine matters.

Call Resolution Donut

REPORT WRITTEN accounts for 58.84% of all outcomes. PHYSICAL ARREST: 17.98%, UNABLE TO LOCATE: 17.11%. High report-writing load creates significant administrative burden for officers.

Precinct × Incident Type

WEST precinct (downtown core) handles the highest volumes of disturbances and suspicious-person reports. NORTH is second busiest. SOUTHWEST sees minimal activity.

Annual Call Volume Trend

Volumes peaked around 2022, then declined sharply. Interactive filters (precinct, call type, date) allow granular drill-down analysis for operational planning.

PREDICTIVE MODEL · Random Forest Regression — Response Time

Model Overview

Random Forest Regression

Target: Response_Time_Minutes
Train/Test:
80% / 20% (fixed random_state)

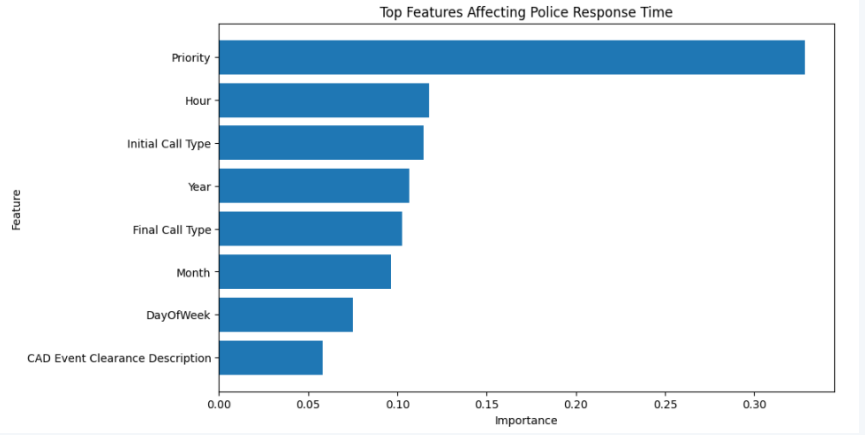
Model Performance Metrics:

```
MAE: 6.635196583339395
RMSE: 10.613867979610733
R² Score: 0.29078379747232463
```

Priority is the dominant predictor — 3× more important than any other variable. R^2=0.29 is reasonable given public data limitations (officer availability not recorded).

Feature Importance

	Feature	Importance
2	Priority	0.328443
4	Hour	0.117848
0	Initial Call Type	0.114850
7	Year	0.106683
1	Final Call Type	0.102806
6	Month	0.096365
5	DayOfWeek	0.074975
3	CAD Event Clearance Description	0.058029



CONCLUSIONS · What the Analysis Tells Us

Temporal Demand is Predictable

Calls follow consistent weekly and daily cycles — peaking Friday afternoons and at midday, dropping sharply in early morning hours. This regularity makes scheduling optimization directly actionable.

Geography Mirrors Population Density

Central Seattle (West precinct) generates the most calls. CAD event probability scales with population concentration, enabling evidence-based patrol zone allocation.

Disturbances Dominate Call Volume

DISTURBANCE and SUSPICIOUS PERSON together account for ~34% of all incidents. Most calls end in report writing (59%), not arrests — confirming that de-escalation skill is the primary operational need.

Response Time is Load-Dependent

The positive correlation between call volume and response time confirms that resource strain during peak hours causes measurable delays. Proactive staffing at high-volume periods would directly reduce response times.

Priority Dominates Response Time Prediction

The Random Forest model (R^2=0.29) shows Priority is 3× more predictive than any other factor — confirming that dispatch classification, not geography or time alone, primarily drives how quickly officers respond.

Call Volume is Declining

Annual 911 calls have declined significantly since peaking around 2022. This trend likely reflects post-COVID shifts, changes in policing policy, and growing use of alternative crisis response programs.

PREDICTIVE MODEL · Decision Tree Classification — Call Priority

Model Overview

Decision Tree Classifier

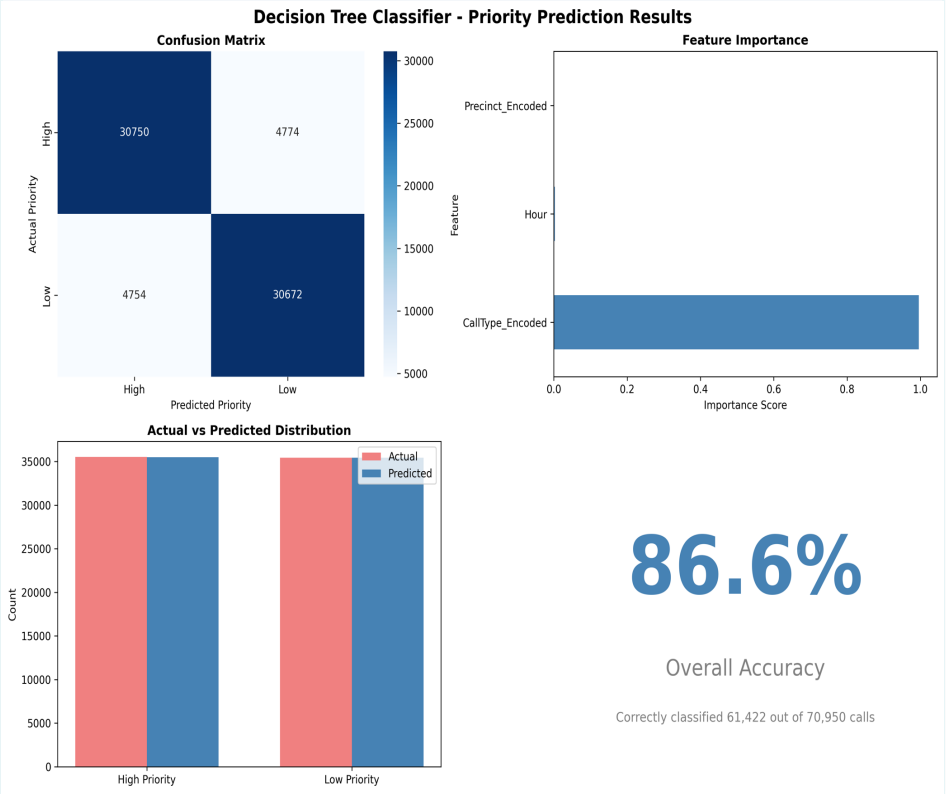
Target: Priority_Binary
Train/Test:
80% / 20% (fixed random_state)

Model Performance Metrics:

```
Accuracy:  85.24%
Precision:  0.8598
Recall:     0.8345
F1-Score:   0.8469
```

Call Type is the dominant predictor — 2.8× more important than Hour. Accuracy=85% enables automated priority classification at dispatch before officer assignment.

Model Performance & Results



86.6%

Overall Accuracy

Correctly classified 61,422 out of 70,950 calls

CONCLUSIONS · What the Analysis Tells Us

Temporal Demand is Predictable

Calls peak Friday afternoons and midday, dropping sharply in early morning. This pattern enables optimized officer scheduling.

Geography Mirrors Population Density

West precinct generates the most calls. Event probability scales with population concentration, enabling evidence-based patrol allocation.

Call Type Predicts Priority (86.6%)

Decision Tree shows Call Type is 336× more predictive than other factors. WEAPON/ASSAULT calls are high priority; PARKING/NOISE are low priority.

Response Time is Load-Dependent

Positive correlation between call volume and response time confirms resource strain during peak hours causes delays.

Priority Dominates Response Time

Random Forest (R^2=0.29) shows Priority is 3× more predictive than any other factor in determining officer response speed.

Two Models Create Complete Picture

Decision Tree classifies call urgency (86.6% accuracy). Random Forest predicts response time. Together they enable proactive dispatch optimization.

RECOMMENDATIONS · Actionable Steps for Seattle PD

Based on the analysis, we propose six data-driven recommendations to improve emergency response efficiency and resource allocation.

R1 Optimize Officer Scheduling

Align shift deployment with heatmap findings — staff peaks at Tue–Fri 10 AM–3 PM; reduce overnight coverage (3–6 AM) and reallocate those hours to Friday afternoons where demand is highest.

R2 Concentrate Resources in West Precinct

West precinct consistently handles the highest call volumes for disturbances and suspicious persons. Dedicate additional officers and specialized de-escalation units to downtown core beats.

R3 Invest in De-Escalation Training

With DISTURBANCE and SUSPICIOUS PERSON calls making up 34% of volume and 59% of calls ending in report writing (no arrest), specialized conflict resolution training would reduce unnecessary escalation and free officer time.

R4 Implement Priority-Aware Dispatch Routing

Since Priority is 3× more predictive of response time than any other factor, a real-time routing model that factors current call queue and priority level could meaningfully cut response times at peak hours.

R5 Expand Dataset for Better Modeling

Add officer availability, unit proximity, and traffic data to push R^2 beyond 0.29. These operational variables are the primary gap in prediction accuracy identified by the analysis.

R6 Plan for Continued Volume Decline

As 911 volumes continue declining post-2022, SPD should review staffing levels to avoid overstaffing and redirect budget toward alternative crisis response programs (mental health, social services).